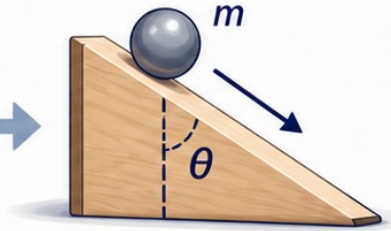
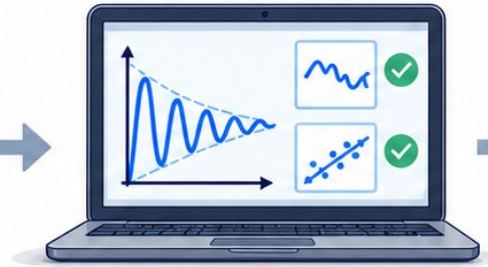
PHY4605 Capstone Investigation

Each group is assigned one project.

Each group investigates its assigned physical scenario, builds a defensible simulation, and supports the conclusion with evidence.









ASSIGNED BRIEF

Scenario, objective and starting boundary

PHYSICAL MODEL

Assumptions, variables, units and scope

SIMULATION + EVIDENCE

One principal output and two checks

DEFENCE + HANDOFF

Explain, rerun and reproduce

The aim is not the most complicated model. The aim is a model another student can check and rerun.

Your Assigned Investigation

Work within the assigned brief; make the scope defensible.

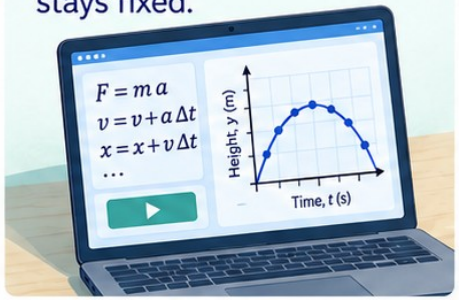
1 UNDERSTAND

State the physical question, objective, assumptions, variables and units.



2 SIMULATE

Use the supplied computational route; make one controlled modification or implement one realism feature, and state what stays fixed.



3 SHOW EVIDENCE

Present one principal graph or table, the required validation check, one different chosen check, and the physical interpretation.



4 DEFEND + HAND OFF

Trace a relevant code section, answer one follow-up, and leave the files, settings and run order reproducible.



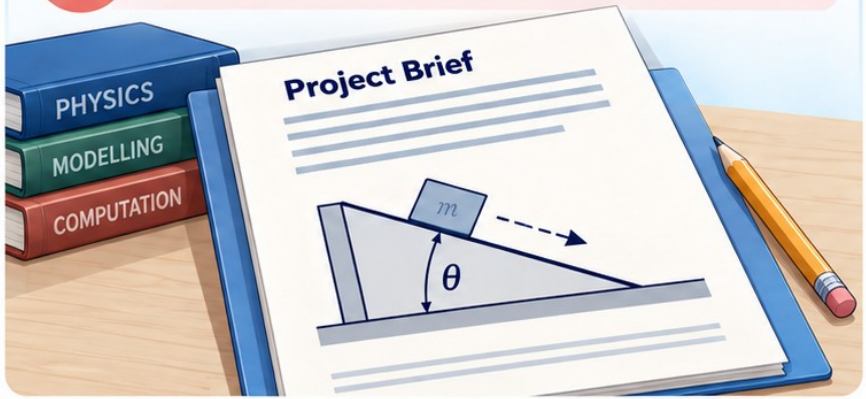
Assigned scenario → model and units → simulation → checks → interpretation → reproducible handoff

Every member must understand the model, output and evidence.

The Brief Sets the Starting Boundary

THE ASSIGNED BRIEF PROVIDES

- Physical scenario and broad objective
- Starting information or data categories
- Relevant setting, process or operating context



BOUNDARY

THE GROUP ESTABLISHES

- Assumptions, variables and units
- Defensible model scope and computational route
- Evidence, interpretation and limitations

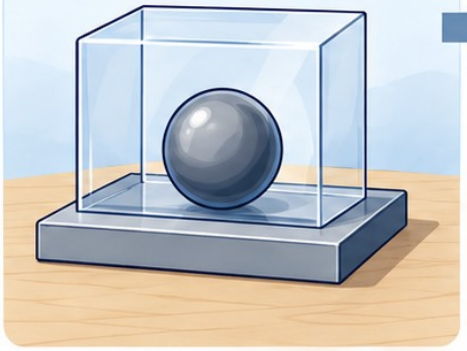


The assigned brief starts the investigation; the group makes the scope defensible.

Realism Means Relevance and Evidence

1 START WITH A BASELINE

Build a transparent, useful model that answers the assigned objective.



2 ADD WHAT MATTERS

Include effects that can materially change the result and can be supported or tested.



3 CHECK THE BEHAVIOUR

Compare with a reference, limiting case, convergence trend or genuinely different check.



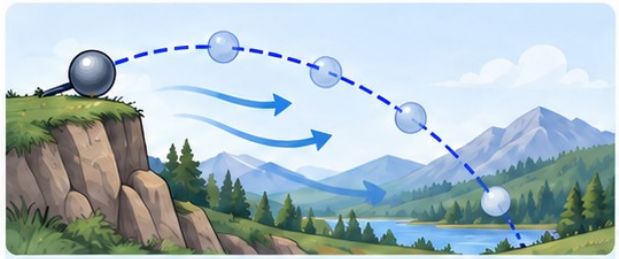
4 DEFEND THE SCOPE

Explain important omissions and separate numerical limits from model limits.



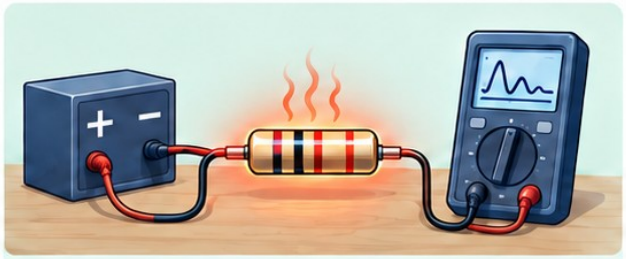
More terms do not automatically mean a better simulation.

Twelve Assigned Physical Investigations



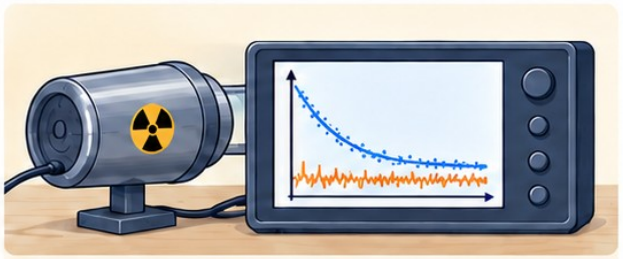
PROJECTILE 01–02

Wind, drag and terrain



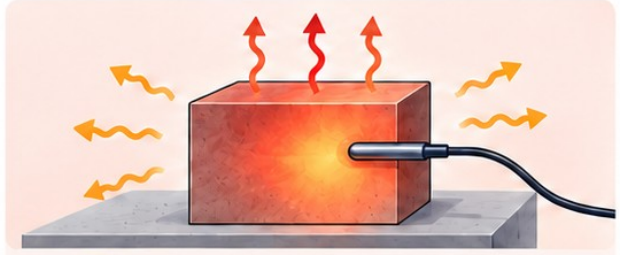
CIRCUIT 03–04

Leakage, temperature and loading



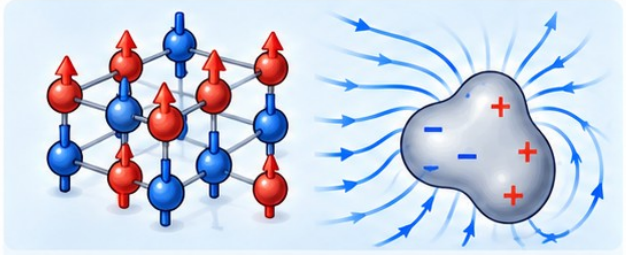
RADIATION 05–06

Background, dead time and decay mixtures



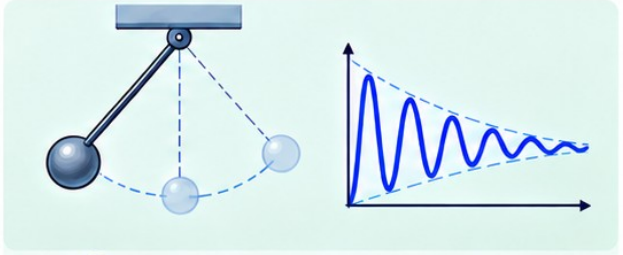
THERMAL 07–08

Radiation, ambient change and internal generation



MONTE CARLO PHYSICS 09–10

Spin lattices and field estimation



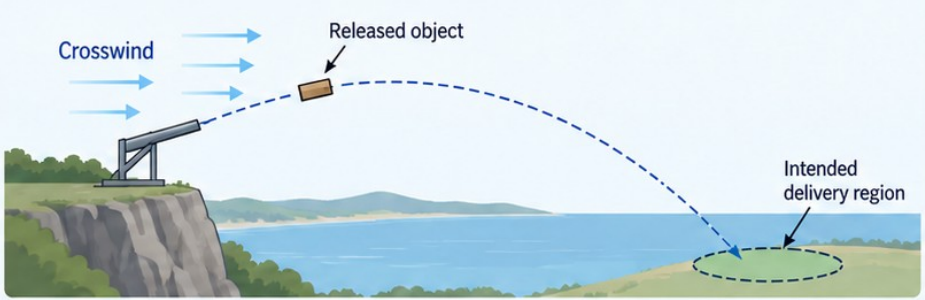
OSCILLATORY 11–12

Damping, finite-angle motion and energy

Each group is assigned one project; the same evidence standard applies to every group.

Briefs 01–02 | Projectile Motion

BRIEF 01 — Projectile Delivery in Crosswind with Quadratic Drag



CONTEXT + OBJECTIVE

A package or test body moves through crosswind toward an intended delivery region. Investigate how convincingly the motion and delivery outcome can be represented.

POSSIBLE REALISM DIMENSIONS

3D motion, relative airflow, varying wind, aerodynamic properties, release uncertainty, air-density variation, lift or spin.

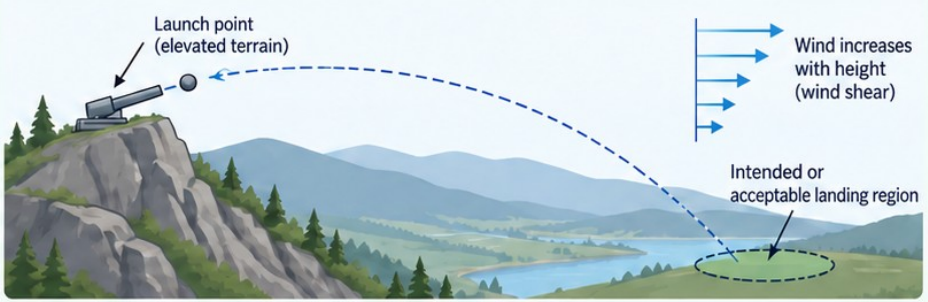
LIKELY INFORMATION

Release conditions, object properties, target region, wind measurements or ranges, and any reference observations actually supplied.

BOUNDARY QUESTIONS

Is a steady crosswind adequate? Is 3D motion material? Can the body be treated as a point? Which aerodynamic effects are supported by evidence?

BRIEF 02 — Projectile Launch from Elevated Terrain under Wind Shear



CONTEXT + OBJECTIVE

An object launches from elevated or uneven terrain while wind changes with height. Investigate whether it can reach an intended or acceptable landing region.

POSSIBLE REALISM DIMENSIONS

Wind shear, terrain profile, non-level impact surface, launch variability, atmospheric variation and local obstructions.

LIKELY INFORMATION

Site elevations or terrain information, wind observations at selected heights, launch conditions, object properties, and an intended landing or safety region.

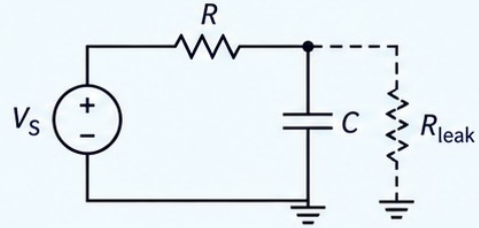
BOUNDARY QUESTIONS

How much terrain is needed? Is 2D defensible? What happens outside measured wind heights? Are gusts, turbulence or obstacles material but unsupported?

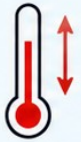
Add effects that materially affect the objective; justify important omissions.

Briefs 03–04 | Circuit Behaviour

BRIEF 03 — Leakage and Temperature-Dependent RC



Temperature affects R, C and leakage



Higher T: larger leakage and/or different R, C values

CONTEXT + OBJECTIVE

A capacitor or sensor node charges or discharges while leakage and temperature alter its time response. Investigate whether a simple time response remains defensible across the assigned operating conditions.

POSSIBLE REALISM DIMENSIONS

Nonzero leakage paths, temperature-dependent resistance or capacitance, source and load resistance, ambient or self-heating, and measurement noise.

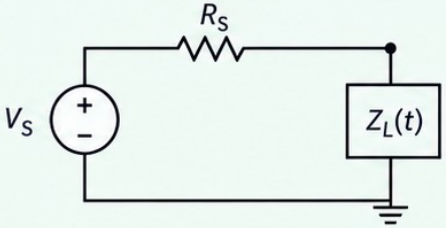
LIKELY INFORMATION

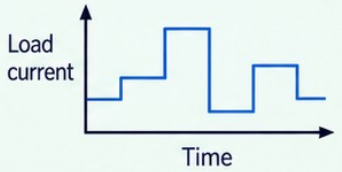
Nominal R and C values, temperature range, leakage or insulation information, initial and final conditions, and measured voltage/current samples or a supplied reference trace.

BOUNDARY QUESTIONS

Is an ideal RC adequate? Which parameters vary with temperature? Is one leakage path enough? When do parasitic or load effects become material?

BRIEF 04 — Internal Source Resistance and a Time-Varying Load





CONTEXT + OBJECTIVE

A real source drives a load whose demand changes with time; internal resistance causes voltage droop and energy loss. Examine the response and whether the supply remains within an acceptable operating region.

POSSIBLE REALISM DIMENSIONS

Internal source resistance, a time-varying load profile, wiring or contact resistance, source limits, thermal effects, and sensor resolution.

LIKELY INFORMATION

Source nominal voltage, internal resistance or loaded/open-circuit observations, load schedule or range, sampling interval, acceptable band, and reference measurements.

BOUNDARY QUESTIONS

Are transients material? Is the load schedule known? Does source sag depend on current or temperature? Is the idealisation defensible outside the measured range?

Include effects that materially change the objective; justify important omissions.

Briefs 05–06 | Radiation and Detection

BRIEF 05 — Background Radiation and Detector Dead Time



CONTEXT + OBJECTIVE

A detector records source events together with background counts, but finite recovery time can miss events at higher rates. Estimate the source contribution and assess how observed counts change with background and event rate.

POSSIBLE REALISM DIMENSIONS

Background rate, detector dead time, acquisition window, detection efficiency, counting randomness, shielding or geometry, and pile-up.

LIKELY INFORMATION

Source and background counts, measurement duration, detector dead time or calibration, geometry or distance, repeated counts, and reference rates.

BOUNDARY QUESTIONS

Can counts be treated as independent? Is dead-time correction material? How is background measured? Is detector efficiency constant? Which rates are supported?

BRIEF 06 — Two-Component Decay Mixture and Counting Variation



CONTEXT + OBJECTIVE

Recorded counts arise from two decaying components with detector efficiency and statistical variation. Determine whether the observed time history supports a two-component interpretation within the available evidence.

POSSIBLE REALISM DIMENSIONS

Two decay timescales, relative initial contributions, efficiency variation, background, finite bin width, and counting noise.

LIKELY INFORMATION

Time stamps or bin widths, counts, efficiency or calibration, background estimates, initial activity or ranges, and reference measurements.

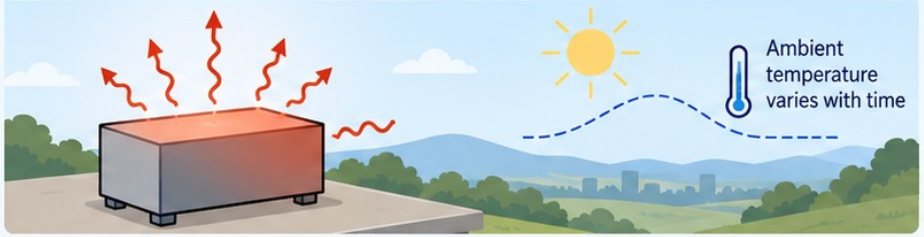
BOUNDARY QUESTIONS

Are two components identifiable? Is efficiency constant? Are intervals long or short relative to decay? How much noise supports a second component? Which alternatives remain unresolved?

Include effects that materially change the objective; justify important omissions.

Briefs 07–08 | Thermal Behaviour

BRIEF 07 — Radiation and a Changing Ambient



CONTEXT + OBJECTIVE

A heated or cooling object exchanges heat by convection and radiation while the ambient temperature varies. Assess whether its temperature history and dominant trend can be represented over the assigned interval.

POSSIBLE REALISM DIMENSIONS

Convection coefficient, radiative exchange, changing ambient, emissivity or surface area, internal thermal mass, and sensor lag.

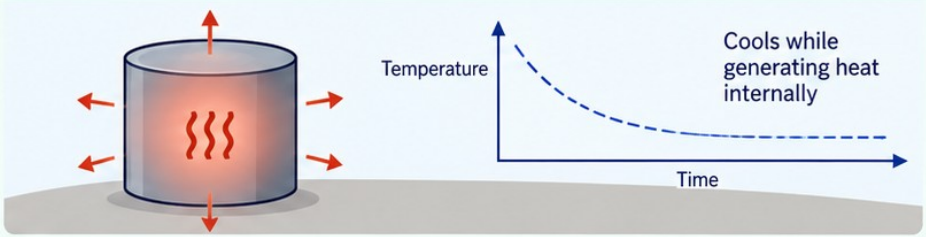
LIKELY INFORMATION

Initial, object, and ambient temperature records, geometry or area, material properties or estimates, airflow or ambient history, and sensor specifications or a reference curve.

BOUNDARY QUESTIONS

Is ambient variation material? Is a lumped temperature defensible? Can radiation be separated from convection? Is sensor lag significant? What lies outside the supported range?

BRIEF 08 — Internal Heat Generation During Cooling



CONTEXT + OBJECTIVE

An object cools while generating heat internally and may approach a nonzero equilibrium. Investigate combined heat loss and generation, and identify conditions for cooling, warming, or steady temperature.

POSSIBLE REALISM DIMENSIONS

Generation-rate variation, spatial gradients, convection, radiation or conduction, changing ambient, material properties, and sensor placement.

LIKELY INFORMATION

Generation schedule or power, initial and ambient temperature, geometry or material information, heat-loss observations, sensor locations, and safe or operating ranges.

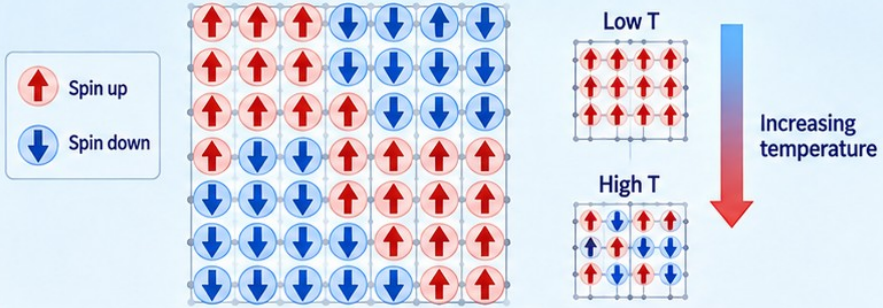
BOUNDARY QUESTIONS

Is a lumped model adequate? Is generation uniform? When does an internal gradient become material? Can losses be inferred from the evidence? Which states are outside the supported range?

Include effects that materially change the objective; justify important omissions.

Briefs 09–10 | Monte Carlo Physics

BRIEF 09 — BOUNDED 2D ISING FERROMAGNET



CONTEXT + OBJECTIVE

A 2D lattice of interacting spins represents a simple ferromagnet. Investigate how temperature changes domain structure, magnetisation and energy, and identify what the finite simulation can support.

POSSIBLE REALISM DIMENSIONS

Temperature, lattice size, boundary condition, coupling strength, initial state, equilibration, random seed and finite-size effects.

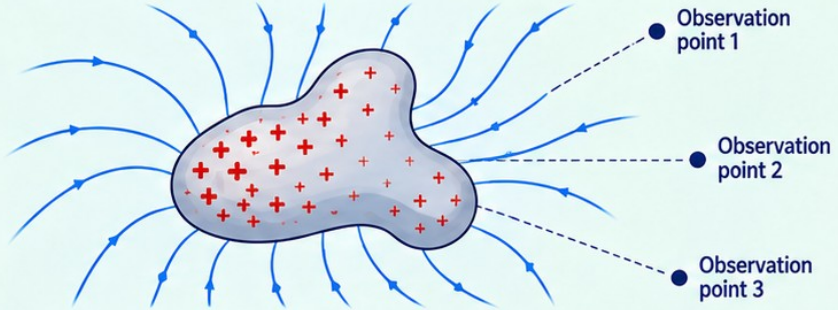
LIKELY INFORMATION

Lattice size, interaction parameter or energy scale, temperature range, supplied or AI-assisted scaffold, initial state and any reference behaviour actually supplied.

BOUNDARY QUESTIONS

How many updates are enough? Is the lattice large enough? Does the initial state matter? Are edge effects material? Is a sharp transition claim defensible?

BRIEF 10 — FIELD OF AN IRREGULAR CHARGED BODY



CONTEXT + OBJECTIVE

A charged body has irregular shape and non-uniform charge density. Use random samples to estimate the electric field at selected observation points and compare the result with a simpler reference case.

POSSIBLE REALISM DIMENSIONS

Geometry, charge-density variation, near-field behaviour, observation-point location, sample distribution, sample count, and singular or excluded regions.

LIKELY INFORMATION

Body geometry or bounding region, charge-density description, observation points, unit and scale information, sample counts and any symmetric reference case actually supplied.

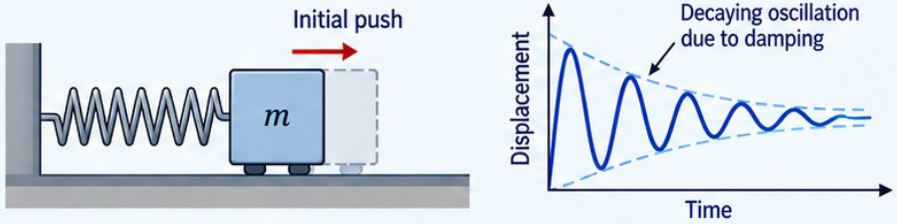
BOUNDARY QUESTIONS

Is the sampling distribution unbiased? How does sample count affect spread? How close can a point be to the source? Is a 2D approximation defensible? When is the estimate converged?

Make each random trial physically meaningful; report mean, spread, sample-size effect and a reproducible check.

Briefs 11–12 | Oscillatory Motion

BRIEF 11 — Damped Mass–Spring Response



CONTEXT + OBJECTIVE

A spring–mass system is disturbed by a short push or changing support and then moves while energy is dissipated. Investigate how one physical feature changes the transient response and settling behaviour.

POSSIBLE REALISM DIMENSIONS

Velocity-dependent damping, pulse or step forcing, nonlinear stiffness at large displacement, moving support, changing mass, and displacement or safety limits.

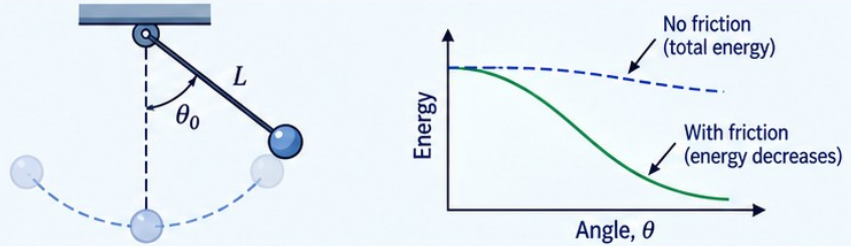
LIKELY INFORMATION

Mass, spring stiffness or baseline response, damping estimate or range, initial position and velocity, force or support history, time window, and an acceptable displacement or settling criterion.

BOUNDARY QUESTIONS

Is one-dimensional motion defensible? Is a linear spring adequate? Is constant damping adequate? Does forcing frequency matter? Is a large timestep hiding phase or energy error?

BRIEF 12 — Finite-Angle Pendulum Motion



CONTEXT + OBJECTIVE

A pendulum is released from a non-small angle and may lose energy at the pivot or to air. Assess when the small-angle picture remains defensible and how one physical feature changes its motion.

POSSIBLE REALISM DIMENSIONS

Finite angle, pivot friction, air resistance, changing length or support vibration, distributed mass or rod inertia, and two- versus three-dimensional motion.

LIKELY INFORMATION

Length, bob or rod mass and geometry, initial angle and angular speed, damping estimate or bounds, gravitational acceleration, time window, and a reference period or observed turning points.

BOUNDARY QUESTIONS

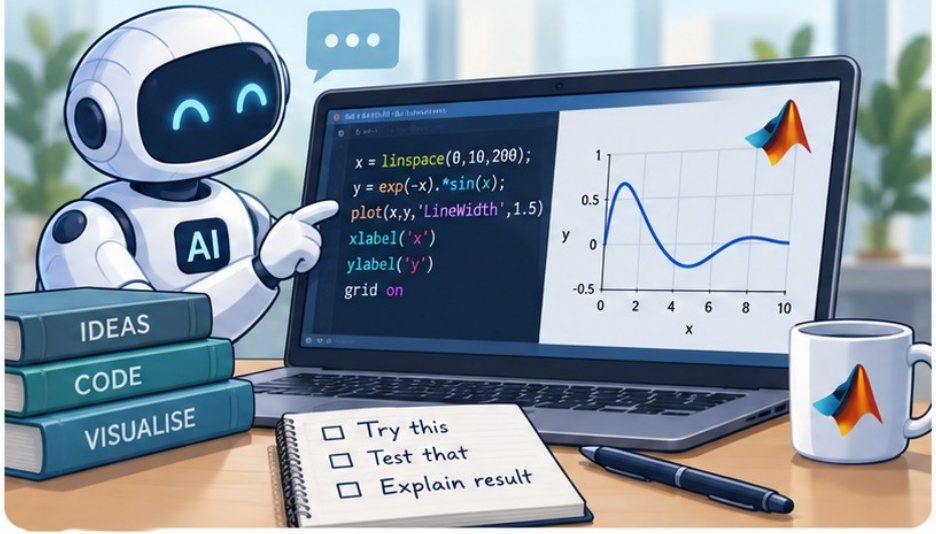
When does the small-angle approximation fail? Is a point mass or rigid rod adequate? Is the damping law supported? Are energy or phase errors numerical or physical? Does a two-dimensional model capture the motion?

Implement one effect that changes the objective; justify important omissions.

AI: Help Build It, Then Verify It Yourself

AI MAY HELP

- Interpret the assigned scenario and clarify the task
- Draft, debug or refine MATLAB code
- Suggest plots, tests or ways to explain the result



THE GROUP MUST

- Inspect and run the final work from a fresh session
- Check units, physical behaviour and numerical evidence
- Verify with a reference, limit, convergence, residual or physical check
- Record what was accepted, changed or rejected, and why



AI output is not evidence until the group runs it, checks it and defends it.

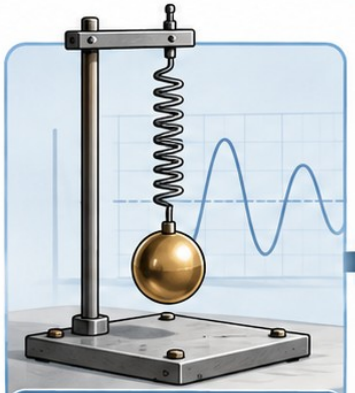
Capstone Rubric | 100%

Seven criteria • four performance levels • total within this rubric: 100%

Criterion	Weight	Excellent	Good	Satisfactory	Weak
Model, assumptions, scope, variables & units	**15%**	Correct, complete, justified, consistent	Appropriate; mostly correct	Basic; partly explained	Missing or physically wrong
Algorithm & MATLAB implementation evidence	**25%**	Clear flow; faithful code; traceable	Mostly correct; traceable	Partial flow; small repair	No code–algorithm link
Principal output & reproducibility	**15%**	Relevant, labelled; fresh-run repeatable	Readable; fresh-run supported	Output; steps incomplete	Missing or not reproducible
Validation evidence	**15%**	Two appropriate, independent checks	Two checks; limited explanation	One valid check shown	No valid check
Physical interpretation & limitations	**15%**	Accurate meaning; consequence-aware limit	Correct trend; relevant limit	Mostly restates; vague limit	Incorrect or overclaims
Presentation & communication	**10%**	Coherent, concise, readable story	Clear story; minor issues	Cluttered; links unclear	Incomplete or hard to follow
Individual defence & AI-use decision record	**5%**	Every member traces; decisions + checks clear	Main path explained; record mostly complete	Partial trace; record gaps	Cannot defend; evidence unusable
TOTAL	**100%**				

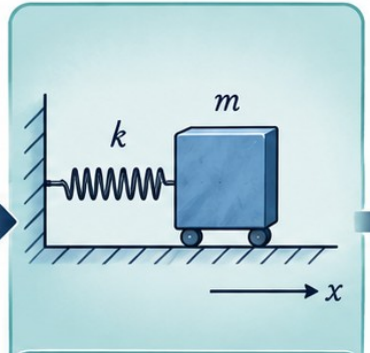
Judge the evidence shown and defended — one rubric, one total of 100%.

Present, Defend and Hand Off



ASSIGNED SCENARIO

State the question, objective and assumptions.



MODEL + REALISM

Explain the scope and the physical feature you included.



SIMULATION + CHECKS

Show the principal output and both validation checks.



INDIVIDUAL DEFENCE

Trace a code section and answer one follow-up.



REPRODUCIBLE HANDOFF

Leave the files, parameters, settings and run order.

Edited script · inputs · parameters + units · numerical settings · checks · conclusion + limitation · AI decision record

Another student should be able to rerun the result from a fresh MATLAB session.